Federico Mariani

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WORK EXPERIENCE

Sefirot Financial Research — **Milano, Italy (Remote)**
Systematic Trading Research Analyst — February 2026 - Present

- Supported quantitative trading at a university association specialized in systematic trading strategies
- Conducted Python-based backtesting of a systematic EUR/USD trading strategy, recognizing key performance drivers and market conditions influencing returns
- Appraised risk-return profile (2:1 ratio, ~45% win rate OOS) and did Monte Carlo simulations to evaluate robustness, identifying max drawdown of about -9% and tail-risk exposure

D+A Strategies — **Dubai, United Arab Emirates (Remote)**
Systematic Trader — August 2025 - February 2026

- Created a systematic trading strategy in an international trading association with 600+ members
- Designed and tested a KAMA-based Bitcoin trading strategy using Bootstrap Grid Search, Bayesian Optimization, and White's Reality Check to assess statistical edge
- Found optimal parameters (Sharpe 0.13, total return 4.6%) and validated results through robustness tests (p-value ≈1), rejecting presence of significant market edge and documenting methodology in LaTeX technical report

Azimut Capital Management — **Ancona, Italy**
Financial Investment Analyst - Intern — June 2025 - July 2025

- Analyzed investment funds of Azimut, an independent asset manager with €145B+ AUM
- Developed a Python-based CAPM model to estimate beta, expected return and alpha using historical data
- Collaborated with financial advisors on asset allocation and client portfolio analysis, contributing to investment recommendations

PROJECTS

Multi-Horizon VIX Forecasting: Econometric, ML and Deep Learning Models with Prediction Market features

- Benchmarked Random Walk, ARIMA, HAR-Type, Random Forest, XGBoost, LSTM and TFT models to forecast the CBOE VIX across three horizons (1, 5, 22 days), valuing point-level and directional accuracy using Python
- Engineered Polymarket (prediction market) features and optimized ML/DL models via Grid Search CV and Bayesian Optimization across multiple VIX forecast horizons, achieving directional accuracy up to 67.86% and discovering TFT as the most competitive Deep Learning architecture for point-forecast accuracy at longer horizons
- Developing a paper based on this thesis

Supervised Machine Learning for Credit Default Prediction

- Established an end-to-end credit risk model on German Credit dataset (1,000 observations, 20 variables), analyzing key financial and demographic drivers of creditworthiness using R
- Built and optimized a logistic regression model with stepwise selection, reducing AIC from 730 to 704 (-3.6%) and residual deviance by ~26%, improving model fit and predictive reliability
- Measured model performance (AUC = 0.78), engineered features (log and polynomial transformations) and documented methodology in LaTeX technical report

EDUCATION

University of Macerata — **Macerata, Italy**
Bachelor's Degree in Banking, Financial and Insurance Economics — December 2022 - October 2026

- **Relevant Coursework:** Financial Analysis, Investment and Asset Management, Financial Mathematics, Statistics, Data Mining

Kozminski University Business School — **Warsaw, Poland**
Erasmus Exchange in Quantitative Finance & Risk Management — September 2025 - February 2026

- **Relevant Coursework:** Statistics for Machine Learning, Derivatives Valuation and Advanced Risk Management, Portfolio Analysis, Corporate Finance, Mergers and Acquisitions, Modern Banking Systems

SKILLS, ACTIVITIES & INTERESTS

Languages: Italian (fluent), English (advanced)
Technical Skills: Excel (advanced), PowerPoint, Python, R, LaTeX
AI Tools: Codex
Interests: Hiking, Entrepreneurship, AI